

Project Summary

Overview:

The objective of this project is to establish technical feasibility of a wireless smart IoT monitoring system to detect, quantify, and report sudden events on civil infrastructures, such as bridge strikes and overweight vehicles. In particular, thousands of these events occur annually on the over 700,000 highway and railroad bridges in the US. Due to the unpredictable nature of such events, many of these events go unnoticed or unreported. However, they can cause structural failures or hidden damage that accelerates the gradual degradation, resulting in over \$200 million annually worth of damage to public and personal property. The proposed method provides a cost-effective wireless IoT solution for bridge monitoring. When deployed on in-service bridges, the potential dangers due to sudden events will be assessed and reported in real-time, enhancing the safety of structures, and mitigating one of the largest loss categories for bridge owners: extended maintenance and downtime costs. The laboratory experiments conducted under this project will demonstrate that the proposed system is commercially feasible and meets documented customer needs.

Key words: structural health monitoring, wireless smart sensors, civil infrastructure, sudden event monitoring, smart trigger, time synchronization.

Subtopic Name: IoT1. IoT Sensors and Actuators.

Intellectual Merit:

This Small Business Innovation Research Phase I project seeks to develop a wireless IoT monitoring system using a network of demand-based multimetric smart sensors working collaboratively. The key technological requirement is to maintain always-on monitoring, so as not to miss an event, with minimal power budget and without sacrificing high-fidelity measurement. However, the potential for realizing the vision of low-cost, ubiquitous sensing is low, primarily because the events of interest are unpredictable and short-duration. Traditional wireless smart sensors (WSS) are poorly suited to detecting sudden events because of duty-cycle employed to preserve limited battery power. Embedor's proposed method would overcome this hurdle by developing demand-based wireless smart sensor network (WSSN). In each WSS, a programmable event-based switch is developed utilizing a low-power triggering accelerometer, and the switch is integrated with a high-fidelity sensor platform. This unique switch would enable the system to maintain ultralow-power operation indefinitely, and after the sudden event is detected, to rapidly transition to high-fidelity measurement. The entire process requires minimal power budget but doesn't lose any data about the event itself, which is a major drawback of previous WSS-based solutions for sudden event monitoring.

The Phase I research objective is to demonstrate the feasibility of the approach in the lab to address the challenges of sudden event monitoring. This demonstration will involve developing a proof-of-concept prototype system and evaluating its ability to detect sudden events with the device initially powered off, and without losing any information about the event. Phase II will focus on transitioning from a laboratory prototype to a field-deployable commercial prototype for testing on in-service bridges.

Broader Impacts:

The commercial potential of smart sensor technology for sudden event monitoring is significant for infrastructure owners and companies providing engineering and maintenance services to them. In particular, state DOTs and railroads have a strong profit motive in the early detection and repair of damage due to sudden events, because the failure of a single bridge can affect the entire transportation system in the community, resulting in large daily economic losses up to one million dollars. The technology developed can also be used for autonomous monitoring of structures, such as, railroad bridge monitoring during train passing by, space grid structures monitoring under blizzards, and utility poles monitoring under blasts.

The successful development and deployment of this technology will also carry a positive societal impact, offering the potential to make the nation's transportation infrastructure safer. In particular, using a single monitoring system deployed on a single bridge to enable overweight vehicle detection will bring indirect societal benefit of more than half a million dollars annually. While in addition to supporting U.S. technology leadership in SHM and IoT, the program will impact future engineering practice and education in a broad sense. Especially, a multi-disciplinary platform of learning and research will be developed for students and community to better understand the monitoring of sudden events, such as earthquakes and collisions.

1. Elevator Pitch

We are developing a novel cost-effective wireless IoT monitoring system to enable rapid condition assessment of civil infrastructures under sudden events. Our initial target customers are bridge owners and companies providing maintenance services to them, because bridges are critical infrastructures which are distinctly vulnerable to sudden events like bridge impacts, overweight vehicles and natural disasters. Currently, each of these events is monitored using different systems which are either costly or inefficient. Our cost-effective system can replace all these monitoring systems, providing actionable inspection information in an efficient manner in advance of the catastrophic failures. The ability for rapid condition assessment would also provide potential societal value by reducing economic losses due to traffic disruption. The key technological innovation of the proposed system is to maintain ultralow-power operation of the wireless system indefinitely, so as not to miss an event, and after the sudden event is detected, to rapidly transition to high-fidelity measurement, with minimal power budget and without losing any information about the event. To date, most traditional wireless monitoring system can't realize this goal.

The proposed system will benefit bridge owners like railroad companies and state DOTs, and companies providing maintenance services of bridges. 700,000 highway and railroad bridges are currently in operation as the lifelines of US commerce. For example, there is an average of one bridge for every 1.4 miles of railroad track, and taking a single bridge out of service due to catastrophic failure can shut down all transportation along hundreds of miles of track, leading to multi-million-dollar losses daily for the railroad company. To ensure the safety of these bridges, \$2.8 billion are spent annually for bridge inspection in the United States. In particular, costly bridge monitoring systems are implemented to detect and mitigate sudden events, such as bridge strikes and overweight vehicles, which are common occurrences and represent leading causes of bridge failures and service interruptions. Especially, to detect overweight vehicles, the market for weigh-in-motion monitoring systems is growing at a significant pace, which was valued at \$512.3 Million in 2017 and is estimated to reach \$875.7 Million by 2023, at a CAGR of 8.88% during the forecast period.

The proposed system for sudden event monitoring is enabled by the patented smart sensor technology developed by Embedor's founders at the University of Illinois. This cost-effective system can replace three different monitoring systems, including bridge strike monitoring, weigh-in-motion, and autonomous monitoring. In particular, it will disrupt the status quo in the bridge inspection market in two ways. First, the installation and operating costs of smart sensor-enabled monitoring systems will be more than 50 times lower than comparable systems in use today (e.g., the SHM solution from Digitexx Data Systems, Inc.), providing immediate up-front savings to infrastructure owners looking to instrument a bridge. Second, the dramatically lower cost and short installation time enable deployment on structures that have remained uninstrumented, opening up a new, currently-unserved segment of the SHM market. In addition, the ability for rapid condition assessment would provide huge societal value by reducing traffic disruption/congestion, thereby significantly lowering daily economic loss which can be up to one million dollars. Moreover, it offers the potential to make the nation's transportation infrastructure safer.

The key technological challenge of sudden event monitoring is to maintain low-power operation of the wireless system indefinitely, while rapidly transitioning to a high-fidelity measurement upon a triggering event, without losing any data about the triggering event itself. To overcome this hurdle, a programmable event-based switch will be developed using an ultralow-power triggering sensor to continuously monitor structural response. This switch is then integrated with a high-fidelity WSS platform and can be activated or deactivated by the triggering sensor based on user-defined threshold of vibration. The proposed monitoring system will finally provide high-fidelity synchronized measurement, with minimal event detection error.

In summary, the goal of this project is to develop a novel cost-effective IoT solution to enable sudden event monitoring of civil infrastructure, with a particular focus on bridges. The proposed solution would significantly benefit bridge owners and companies providing bridge maintenance services by reducing inspection cost of monitoring systems, lowering daily economic losses due to traffic disruption, and ensuring the safety of transportation systems.

3. The Innovation

3.1 Brief Description of the Innovation and Proposed Development

A novel commercial prototype wireless smart sensor (WSS), called demand-based WSS, is being developed (See Figure 1), with the unique capability to rapidly acquire massive amounts of high fidelity sensor data at high sampling rates at the onset of a sudden event. The key technological challenge is to maintain ultralow-power operation of the wireless system indefinitely, so as not to miss an event, and after the sudden event is detected, to rapidly transition to a high-power state for distributed data acquisition, without losing any data about the event itself. To overcome this challenge, a programmable event-based switch has been developed using an ultralow-power triggering accelerometer to continuously monitor structural response. This switch is then integrated with a high-fidelity WSS platform; and it can be activated or deactivated based on user-defined threshold of acceleration. If an event occurs while the platform is powered off, the switch turns on the platform to start high-fidelity measurement immediately; if the platform is busy with other tasks, the switch resets the platform and forces it to enter the sensing state. Moreover, when the event stops, the switch interrupts high-fidelity sensing and turns off the platform. Subsequently, in real time, the data is analyzed at the local wireless network level to determine the location and severity of the event, as well as to propose a recommended course of action to report to the infrastructure owner.



Figure 1. The solar-powered demand-based Smart Sensor supports high-fidelity distributed sensing, while featuring a low-power operating mode that enables long-term operation.

Preliminary investigations conducted by Embedor found candidates for such low-power triggering devices that can operate continuously for over 5 years using a single lithium battery. In the high-power mode, the Xnode is solar-powered, providing virtually operation on a rechargeable battery. If satisfactorily implemented, a solution employing these devices would meet the indefinite operation requirement outlined above. In addition, the transition from low-power sensing to high-fidelity data acquisition only takes 1s, ensuring the ability of capturing the most of transient structural response under the event. The more challenging aspect is doing so while meeting other requirements: not losing any data of high quality surrounding the onset of a sudden event, and obtaining synchronized sensor data within a network of demand-based WSSs in the aftermath of the event.

3.2 Key Technological Challenges and Risks

Three major technological challenges exist in the development of sudden event monitoring using wireless smart sensors:

- (1) Capturing critical transient response

Most transient response of structures under sudden events can only last for a few seconds, and thereby structural response at the onset of an event is of great importance but extremely difficult to be captured for subsequent analysis. Typical acceleration response under a bridge strike is shown in Figure 2, by employing the proposed event-based switch. Though the event is detected, and high-fidelity data acquisition starts immediately, most critical structural response is still lost.

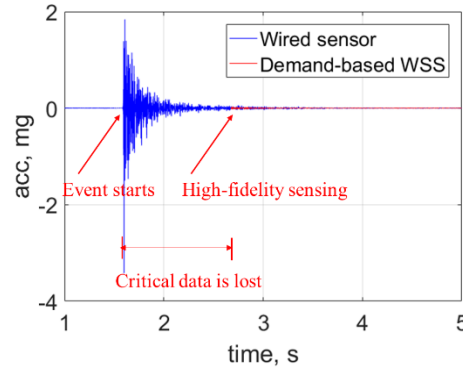


Figure 2. The transient response of a bridge strike.

(2) Synchronizing wireless sensor nodes

In wireless smart sensor network (WSSN) deployed on a structure, each WSS has its own clock which does not share global time. If the analysis is carried out using unsynchronized data, structural condition assessment may be negatively affected, e.g. falsely indicating structural damage. For demand-based WSSN, this issue is more challenging, because the triggering time for each WSS is different and unknown, due to varying vibration levels in the structure at the onset of a sudden event.

(3) Minimizing detection errors

One of important concerns that preclude WSSs for field applications is its reliability in long-term deployment. For demand-based WSSs, the concern of sensor reliability refers specifically to event detection errors. More precisely, it is challenging to achieve 0% false negative rate and minimal false positive rate. In particular, high false positive rate of sudden event detection will result in not only the depletion of battery in a very short period, but also the missing of real events.

3.3 Intellectual Property and NSF Lineage

The innovation proposed by Embedor Technologies would be impossible without a smart sensor platform suitable for structural health monitoring applications. This technology was originally developed by Embedor's founders at the University of Illinois over more than a decade of collaborative interdisciplinary research, including through the support of multiple NSF-funded projects on wireless sensors, structural health monitoring and cyber-physical systems under grants CMS 03-01140, CMS 06-00433, CMMI 09-2888, and CPS 10-35562. That smart sensor technology for structural health monitoring is patent-protected ("High sensitivity environmental sensor board and methods for structural health monitoring," U.S. Pat. 9086430). Dr. Spencer and Dr. Agha, Embedor's co-founders, are among the inventors, while the Board of Directors of the University of Illinois is the assignee of the patent. *Embedor has signed a licensing agreement with the University of Illinois*, which includes the option to exclusive licensing rights once the implementation of the patented technology is ready for commercialization.

The research and development of a wireless IoT monitoring system proposed herein is expected to culminate in a separate patentable innovation, with *the resulting intellectual property being wholly owned by Embedor Technologies*, where the entirety of the R&D activities will take place.

5. Technical Discussion and R&D Plan

5.1 Technical Background

For monitoring bridges under sudden events (e.g. bridge strikes and overweight vehicles), rapid and accurate assessment is critical. An early warning to the bridge owners may be the first step, followed by reports on locations and the severities of the events. Required features for the monitoring systems should also include estimation of potential primary/secondary damage. Thus, a system is needed that can detect, quantify, and report sudden events; however due to the unpredictable nature of sudden events, only a few attempts have been made to date.

Visual inspection has been the common practice in inspecting and monitoring the condition of civil infrastructures after sudden events occur. However, it requires intensive labor and may result in significant delay for emergency response and informed decisions. To overcome these issues, monitoring systems are installed using wired sensors, which is prohibitive for many in large-scale structures due to the significant cost, ranging from \$5K to \$20K per channel, and complicated sensor wiring. In contrast to wired monitoring systems, wireless sensors offer a tremendous opportunity to lower costs and realize the promise of pervasive sensing. However, sudden event monitoring of civil infrastructure using traditional wireless sensors is challenging. For example, WSSs on the Golden Gate Bridge were unable to detect earthquakes that occurred during the three-month monitoring expedition, because of two main issues: (i) wireless sensors are duty-cycled to conserve limited battery power, and may miss sudden events when in power-saving sleep mode; (ii) some sensors are busy with other operations (e.g. data transmission) and have large response latency to detect events.

To date, some progress has been made to the aforementioned challenges. To reduce the response latency, Cheng & Pakzad proposed a pulse-based media access control protocol to timely disseminate trigger message with high priority. Dorvash et al. developed Sandwich nodes to enable interrupting the undergoing tasks for sudden events. Although response delay is reduced in these two strategies, the radio of wireless sensors is always on to keep listening to the messages, resulting in a significant power draw. Moreover, some efforts have been made to extend the lifetime of battery-powered wireless sensors. The TelosW platform employs low-power wake-on sensors and wake-on radios. Similarly, low-power event detector circuit and wake-up receivers are employed in the development of the Zippy technique. Although these two technologies achieve low power consumption, they do not satisfy the high-fidelity requirement of sudden event monitoring for civil infrastructure. For example, the TelosW's analog to digital converter has only 12-bit resolution. Despite these significant advances, to the best of knowledge of the authors, neither a research nor a commercial monitoring system using wireless sensors that can successfully detect, quantify, and report sudden events has been developed. Therefore, an integrated wireless system capable of quickly triggering and seamlessly capturing high-fidelity synchronized data for civil infrastructure under sudden events remains a challenge.

5.2 Team's Prior Work

Embedor's founders have collaborated for over 10 years on NSF-funded research involving wireless smart sensors for structural health monitoring. This research has initially built upon the Imote2 platform, overcoming numerous hardware and software hurdles of full-scale implementations. These developments were showcased in a US-Korea-Japan collaboration that deployed a large-scale long-term deployment of wireless smart sensors for infrastructure monitoring (See Figure 3). Moreover, the NSF-funded research resulted in the Xnode, a commercial prototype wireless smart sensor using patented technology licensed from the University of Illinois. The Xnode has the unique capability to rapidly acquire massive amounts of high fidelity synchronized sensor data from hundreds of sensor channels at high sampling rates, offering

unparalleled capabilities for data acquisition and edge computing—processing data where it's being created, before it reaches the network. Combined with proprietary synchronized distributed sensing capabilities, the Xnode is clearly differentiated from other commercially available wireless sensor and IoT device platforms. Originally conceived for the purposes of long-term periodic structural health monitoring, the Xnode acts as a highly customizable hardware and software platform for a variety of wireless sensing applications, becoming a technology enabler and solution accelerator. The 190 Xnodes have constituted the world's largest deployment of structural health monitoring on Dubai-I Ferris Wheel; the Xnodes can continuously measure the vibration of spoke cables on the rotating wheels, while the conventional wired sensors are very difficult to implement it (See Figure 4).

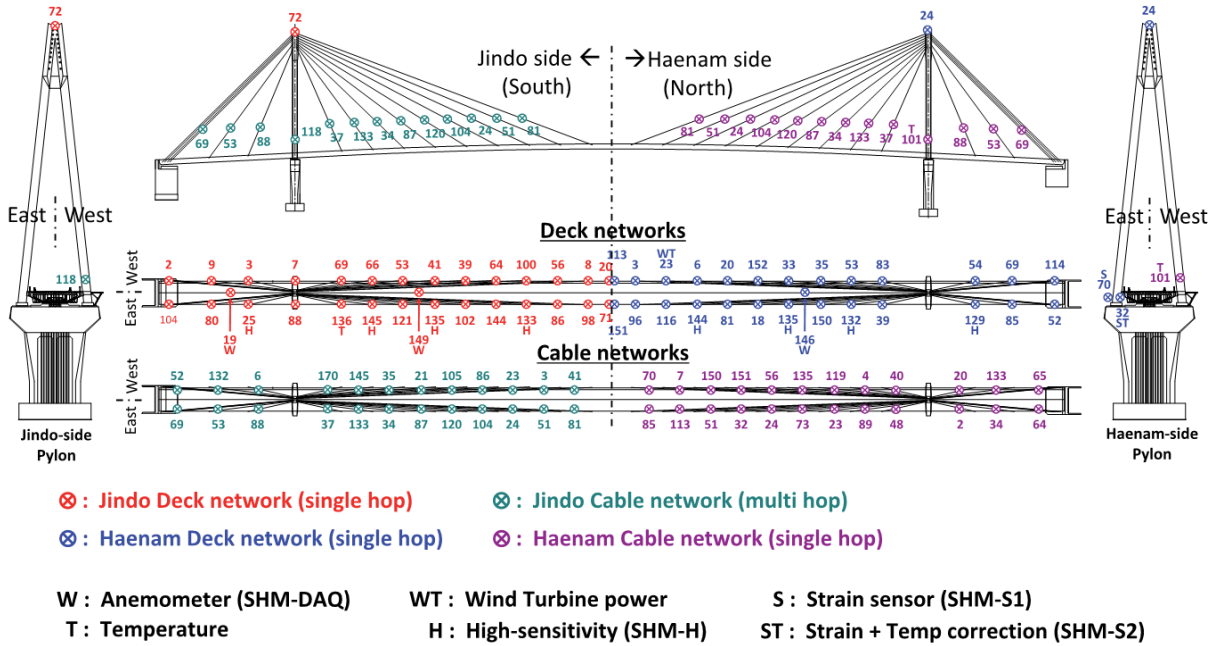


Figure 3. Wireless sensor network deployed on the 2nd Jindo Bridge (Jang et al. 2010). This deployment validated the use of wireless sensors for periodic structural health monitoring of bridges.

Figure 4. Wireless sensor network on Dubai-I Ferris Wheel

5.3 Preliminary Studies

By comparing the low-power accelerometers with trigger in the market, ADXL362 is considered as the best candidate to build demand-based WSSs for sudden event monitoring. In particular, it consumes 15uA at 3.3V or 49.5 μ W, which can continuous work for around 8 years using a single lithium battery; it also has a large FIFO buffer that allows the sensor to save up to 512 samples, which correspond to 1.7s for all three axis sampled at 100Hz surrounding the onset of a sudden event. Moreover, the Xnode Smart Sensor is selected as the high fidelity wireless sensor platform for demand-based WSSs because of its attractive features in powerful hardware and a robust software framework that can simplify the development of sudden event monitoring applications. A preliminary lab prototype of the demand-based WSS is then developed by integrating ADXL362 and Xnode, to validate the potential of event-triggered sensing. The preliminary lab prototype and a wired sensor were both installed on a building model that was mounted on a shaking table. A sudden event is mimicked by short-duration ground motion generated by the shaking table. Figure 5 shows the measured transient response under the sudden event retrieved from the preliminary lab prototype. As can be seen, when the acceleration exceeds the threshold at 6.3s, an interrupt signal is

triggered, and high-fidelity sensing starts after 0.96s. Finally, the most transient responses of the structure is well captured.

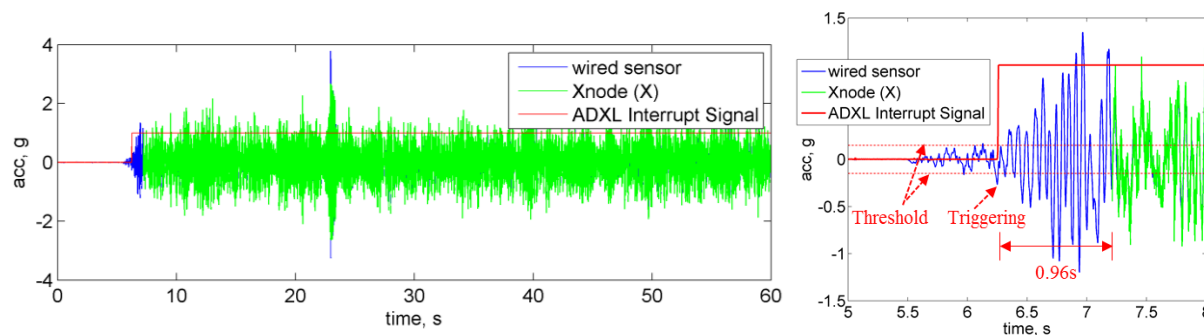


Figure 5. Measured transient responses from a preliminary lab prototype, indicating an amplitude-based threshold triggering mechanism. When the signal amplitude exceeds the threshold, an interrupt is generated, and high-fidelity sensing starts.

5.4 Research & Development Goals and Milestones

The overall goal of this SBIR Phase I effort is to assess the feasibility of the demand-based WSS for commercial deployment on bridges under sudden events. The specific research objectives to achieve are:

1. Design a hardware prototype of the demand-based WSS and the associated software to seamlessly capture critical transient responses without losing any of the triggering event sensor data.
2. Propose a post-event time synchronization protocol to synchronize a network of prototype nodes deployed on a structure in the aftermath of sudden events.
3. Develop strategies of smart adaptive triggering to minimize event detection errors, achieving 0% false negative rate and 5% false positive rate.
4. Conduct laboratory testing of the prototype nodes using a model structure using an instrumented hammer to determine the magnitude of the bridge strike. Demonstrate that the system can obtain high-fidelity synchronized transient responses of this structure.

The critical technical milestones that must be met to for success of Phase I are as follows:

- A working prototype of the demand-based WSS must be developed, with the ability to capture the entire transient response of the structure.
- A post-event time synchronization protocol must be validated using a group of prototype nodes, with a synchronization error of less than 20 μ s.
- Smart adaptive triggering strategy must be developed and tested in long-term deployment, with 0% false positive rate and 5% false positive rate.
- Laboratory test must demonstrate the ability of the prototype to capture high-fidelity synchronized bridge responses during the sudden event.

Completion of these tasks and meeting these milestones will directly support SBIR Phase II efforts designed to lead to a viable commercial product, which include:

1. Conduct field tests at the Transportation Technology Center, Inc. (TTCI), which is the new technology proving grounds for the Association of American Railroads in Pueblo, Colorado (Figure 6).

2. Demonstrate through simulation that information acquired in Phase I is sufficient to perform condition assessment on the full-scale structure after the sudden events (e.g. bridge strikes and overweight vehicles) for a broad range of bridge types.
3. Design and develop an interface to provide actionable information from the system to bridge owners/maintainers.
4. Conduct follow-up field testing on a structure owned by one of the potential customers (e.g., Canadian National), with their engineers as end-users of the system, and collect feedback.
5. Address environmental and durability concerns using realistic field testing, as well as usability and missing or extraneous features of the system from end-user feedback.

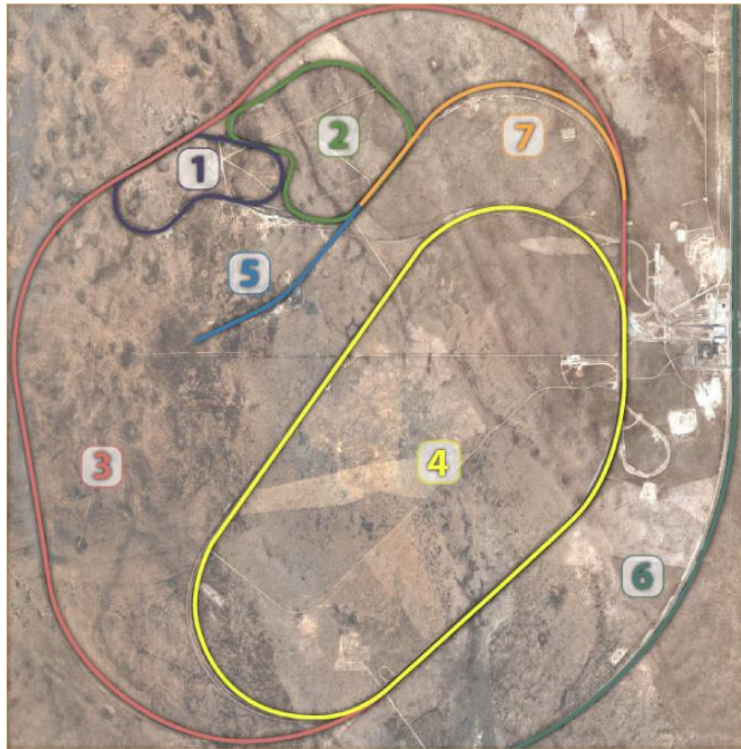


Figure 6. The seven TTCI test tracks in Pueblo, Colorado provide for a variety of technology testbeds, including multiple bridge testing sites, to evaluate innovations of interest to the bridge owners. Image credit: Transportation Technology Center, Inc.

5.4 Scope of Work

Building on the substantial research results obtained and the infrastructure developed by the Embedor team members at the University of Illinois under previously funded NSF grants, four high-priority tasks are proposed in the feasibility assessment of the wireless IoT monitoring system using demand-based WSSs for sudden event monitoring. Note that certain aspects of these tasks are independent and will be pursued in parallel.

Task I: Design prototype of the demand-based WSS to capture entire transient response.

The potential of event-triggered sensing using preliminary prototype has been demonstrated in Section 5.2. However, certain limitations have been realized from the test result. More specifically, as illustrated in Figure 2 and Figure 5, the preliminary prototype is unable to capture the first second of transient response which may be of the utmost importance for subsequent analysis. The 512 samples of FIFO data from the

ADXL362 is considered to be integrated with high-fidelity sensor data from Xnode. To realize this idea, two hurdles must be addressed: (i) clock frequency variability is large in ADXL362, resulting in large variability of data rate, and (ii) variance in the start-up sensing time in Xnode leads to time offset between the two data sources. To overcome these hurdles, on-board post-sensing data fusion is proposed as shown in Figure 7, in which the data of first 0.6s in Xnode is utilized to calibrate the data of last 0.6s in ADXL362. The estimated calibration parameters, including sampling frequency (f_s) and time offset (T), are applied to adjust the entire ADXL362 data.

Dr. Mechitov will lead work on this task. The success criterion for this task is successful realization of a working prototype to capture the entire transient response under sudden events.

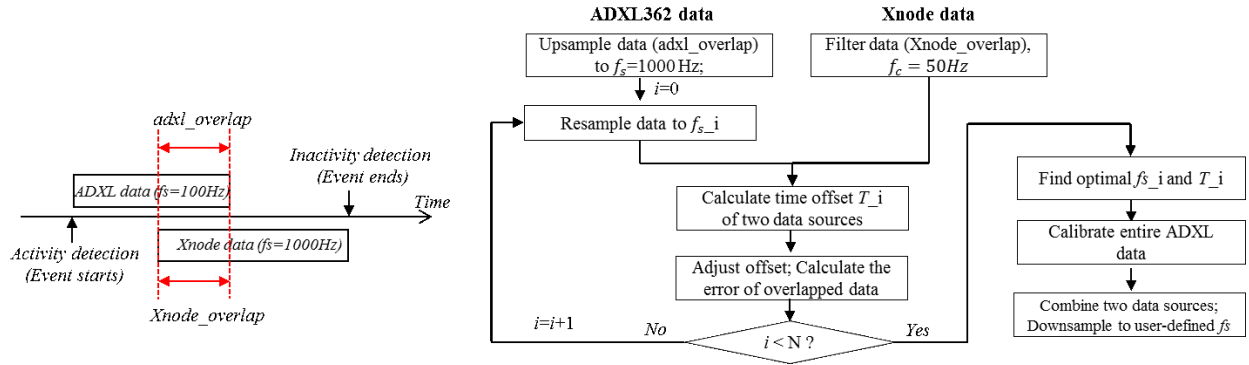


Figure 7. Flowchart of post-sensing data fusion

Task II: Propose a post-event time synchronization protocol to synchronize a network of prototype nodes

Considering a cluster of prototype nodes deployed on a bridge structure, synchronization between different nodes should be addressed for subsequent analysis. To achieve this goal, post-event time synchronization is proposed and will be implemented (Figure 8), consisting of three main steps: 1) once the nodes start sensing, they record their local clock time; 2) after the event ends, the node which finishes sensing first will broadcast its local time stamps, serving as global time stamps, to all other nodes in multiple times; 3) then, all the other nodes will perform post-sensing time synchronization. Finally, data from different WSSs are started at the same time and sampled at the same frequency.

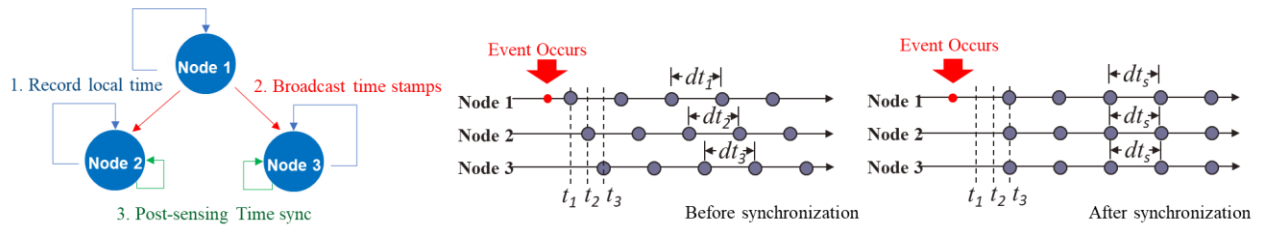


Figure 8. Flowchart of post-event time synchronization

Dr. Spencer will lead work on this task. The success criteria for this task is that minimal time synchronization error of 20us between different nodes is achieved by applying the proposed protocol.

Task III. Develop adaptive smart triggering to minimize event detection errors for long-term deployment

To minimize detection errors, a strategy of adaptive smart triggering will be developed. More specifically, every time an event is detected, the node will determine if it is a real event based on the duration and

amplitude of acceleration measurement. If the event is not real, the measurement data will be thrown away to save memory, and the threshold condition will be adaptively increased to reduce the false positive rate. The strategy will be finally ported to the wireless IoT monitoring system, and validated through long-term lab test.

Dr. Agha will lead work on this task. The success criteria for this task is that event detection error will be minimized to close to 0%.

Task IV: Proof-of-concept testing and evaluation

The final task in this project is to design and conduct thorough proof-of-concept experiments on a laboratory test structure. It is well understood that experimentation is critical to advance toward implementation in full-scale infrastructure systems. Small-scale laboratory tests on scale models must have a central focus on the implementability of the devices and strategies. Such experiments provide insight and expose pitfalls that are not considered in computer simulation studies.

A model structure will be used for laboratory testing. The prototype demand-based WSSs will be mounted on the structure, and an instrumented hammer used to impart an impact of known magnitude. The structure will also be instrumented with wired sensors to serve as a benchmark to compare the proposed system. Data will be collected and analyzed to demonstrate that the system can capture high-fidelity synchronized transient response in an efficient manner.

All team members will collaborate on this task. The success criterion for this task is that good agreement is found between the wired sensors and the wireless IoT monitoring system.

5.5 Schedule

Successful completion of these tasks within the proposed nine-month timeframe for Phase I (Figure 9) will result in the development and operational validation of a wireless IoT monitoring system. Observed performance characteristics and sensor data collected in the laboratory testing of the prototype will play an important role in guiding further research and development activities during Phase II.

	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Task I									
Task II									
Task III									
Task IV									

Figure 9. Gantt chart for the Phase I project tasks. A nine-month timeframe for the completion of Phase I is proposed.